EXAMINATION

Luleå University of Technology

Course: D7024E
Course name: Distributed and Mobile Computing Systems
Date: 2023-10-21
Time: 9.00-13.00, online home exam
Aid: WWW and Generative AI (e.g., ChatGPT, Bard) (no interactive communication with other people)

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Grade scale: 50% grade 3, 75% grade 4 and 90% grade 5	Note that this grade is for the exam. The exam is supposed to be easy enough so that anyone that followed the course properly should pass it. The grade for the course (including all assessment criteria) may be different from the grade of the exam.
Total number of questions and score: 2 questions, total score 36	
Other information: This is an individual home exam.	Note: No interactive cooperation with other people is allowed. Information can be searched on the world wide web, and generative AI (e.g., ChatGPT, Bard, etc.). Please provide references, when such information is used in you answers.

General instructions

You shall formulate answers with your own words and if you use generative AI there must be a personal reflection included. In any case you must provide the references to your primary sources (e.g., URL). Your **solution must be uploaded in the Canvas D7024E room under Assignments**, at the latest at 13:00 (unless you have gotten approval for extra time). **Please put your name both in the filename and on the first line of your document.**

After examination

Examination results are posted on Canvas within 4 weeks after the examination.

1. **Mixed questions on distributed systems (10p)**

   These are multi-choice questions, with three alternatives. Please answer "true", "false" or “depends” to each of the following statements. If your answer is “depends” you must provide a short explanation to get credits. For “true” or “false”, no explanation is required, unless you want to explain what conditions are assumed. If you have identified a corner case and provide a good explanation, we can accept an answer that is different from the expected one. Each question is worth 0.5p

   a. Marshaling (a k a serialization) is used to convert data into a platform dependent format.

   b. In databases there is a need for a local transaction log (or command log) to handle potential loss of RAM cache state, to allow client-defined transactions, and to provide ACID properties.

   c. In databases, deadlocks can be resolved by time-outs and rolling back performed commands.

   d. NOSQL databases typically provide ACID properties across any pieces of data.

   e. Non-Byzantine global consensus solutions typically assume that there can be a small number of traitors.

   f. In the general case, to obtain global consensus you must make assumptions regarding connectivity, or fraction of nodes being alive.

   g. NAT/firewall traversal by TURN enables direct peer-to-peer communication (i.e., eliminating the need for a third-party relay node in the middle)

   h. DNS round robin can advertise several alternative IP addresses for a given URL.

   i. Providing mutable messages is a key paradigm of distributed systems.

   j. Master-slave architectures are often used for scalability where fast and immediate consistency is required.

   k. Rendezvous handshaking is a typical approach in geo-distributed systems.

   l. Document databases typically provide good data locality.

   m. Kafka is completely distributed (no internal global synchronization needed).

   n. Scalable reliable multicast (SRM) can in worst case suffer from acknowledgement (ack) explosions.

   o. For scaling out a system with eventual consistency, performance is improved if we adopt a notion of transactions to support multiple commands atomically.

   p. Safety and liveness must be ensured to reach global consensus.

q. Zookeeper is a good solution for large number of collaborating nodes to record and keep track of whether each of them is alive/reachable or not.

r. Data storage that is based on consistent hashing places data so that short round trip times are obtained.

s. Map-reduce in Hadoop is designed for highly geographically distributed big data analysis.

t. A distributed hash table (DHT) has better scaling properties than immediate consistent hashing (e.g., as used in Cassandra).

2. Message systems, microservices and blockchain (26p)

a) What are the key properties of PubSub and message queues respectively? What are the key properties of the messages therein? (6p)

b) Explain the principles of microservices and how they typically support the event-driven paradigm. What are the key benefits and challenges? (6p)

c) How do online multiplayer real-time games (e.g., FPS games) provide effective and efficient synchronization? Mention the four (4 and not more) most important properties/approaches in your opinion (8p)

d) Distinguish between on-chain and off-chain solution. Explain the operational features of Ethereum sharding and Cardano Hydra. (6p)

Best Wishes!
Finally, please fill out the **online course evaluation**. An email with a link has been sent to you.